

MACHINE LEARNING

INTRODUCTION

NGO QUOC VIET-2025

CONTENTS

1. Introduction to ML
   - What is ML?
   - The history.
   - The concepts, terminologies.
2. The ML examples.
3. The researches and applications in ML.
4. Get started with Regression model.

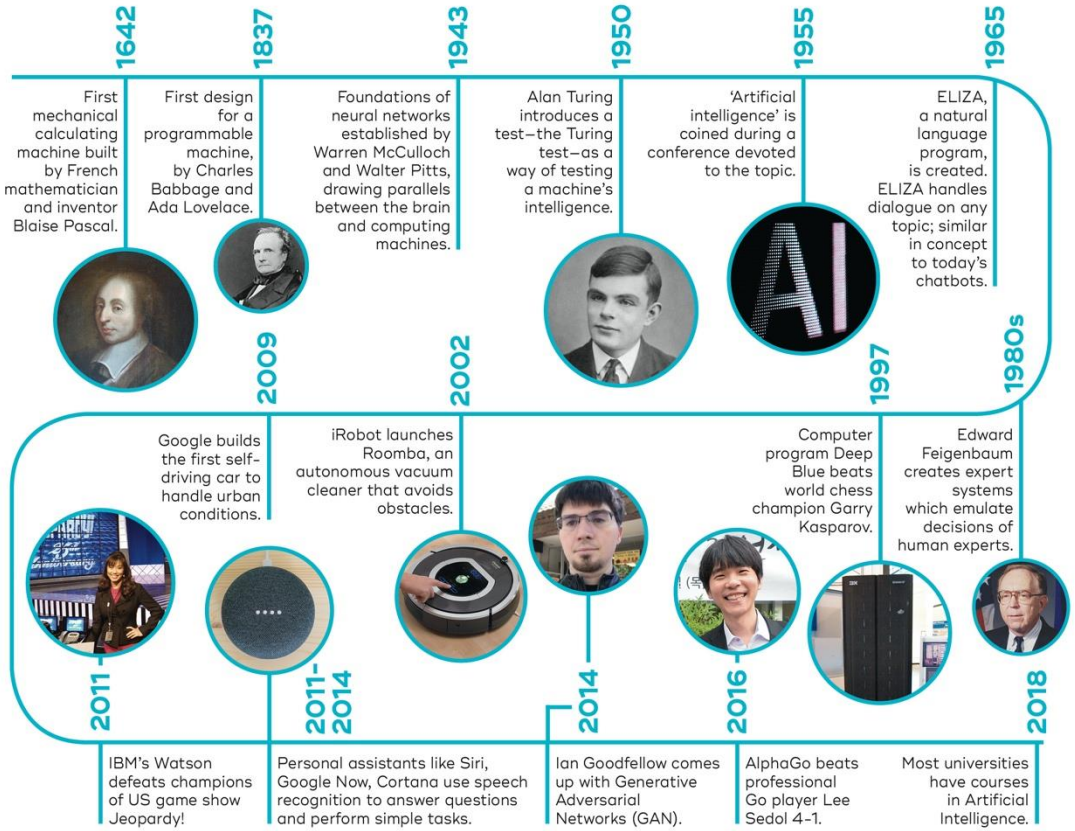
WHAT IS MACHINE LEARNING?

- By Google: The use and development of computer systems that are able to learn and adapt without following explicit instructions, by using algorithms and statistical models to analyze and draw inferences from patterns in data.
- By https://www.computermachinelearn.com: Machine learning is an application of Artificial intelligence (AI) that provides a systems the ability to automatically learn and improve from the experience without being explicitly programmed. **Machine learning focuses on the development of computer programs** that can access data and use it learn for themselves
- By Wiki: **Machine learning** (**ML**) is the study of computer algorithms that improve automatically through experience. It is seen as a subset of artificial intelligence. Machine learning algorithms build a model based on sample data, known as "training data", in order to make predictions or decisions without being explicitly programmed to do so.

THE HISTORY OF MACHINE LEARNING

- Two undergraduate students at Havard, built the first neural network computer in 1950 using 3000 vacuum tubes.
- Alain Turing "Computing Machinery and Intelligence."
  - Turing Test, Machine Learning, Genetic algorithms, and reinforcement learning.
  - Instead of trying to simulate the adult mind, why not try with a child's.

https://www.youtube.com/watch?v=m_nQM_AkVfY



https://qbi.uq.edu.au/brain/intelligent-machines/history-artificial-intelligence

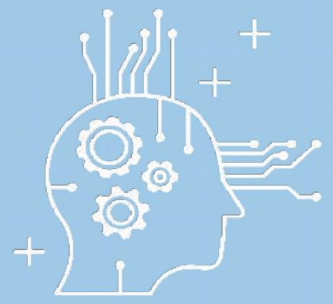
COURSE STRUCTURE

- Classical Machine learning
  - The CCR family (classification,clustering, and
  - regression)
  - Feature engineering, and model tuning
  - VC dimension, and universality theorem
- Deep Learning
  - Vision (CNN)
  - NLP and sequence processing (RNN)
  - Deep RL or GAN (if you're still alive!)

MACHINE LEARNING AS SUBFIELD OF AI

Artificial Intelligence

Any technique that enables computers to mimic human behavior



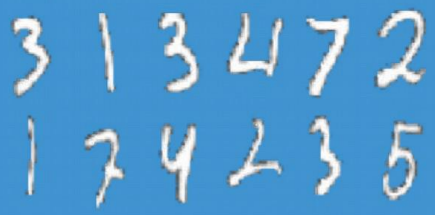
Machine Learning

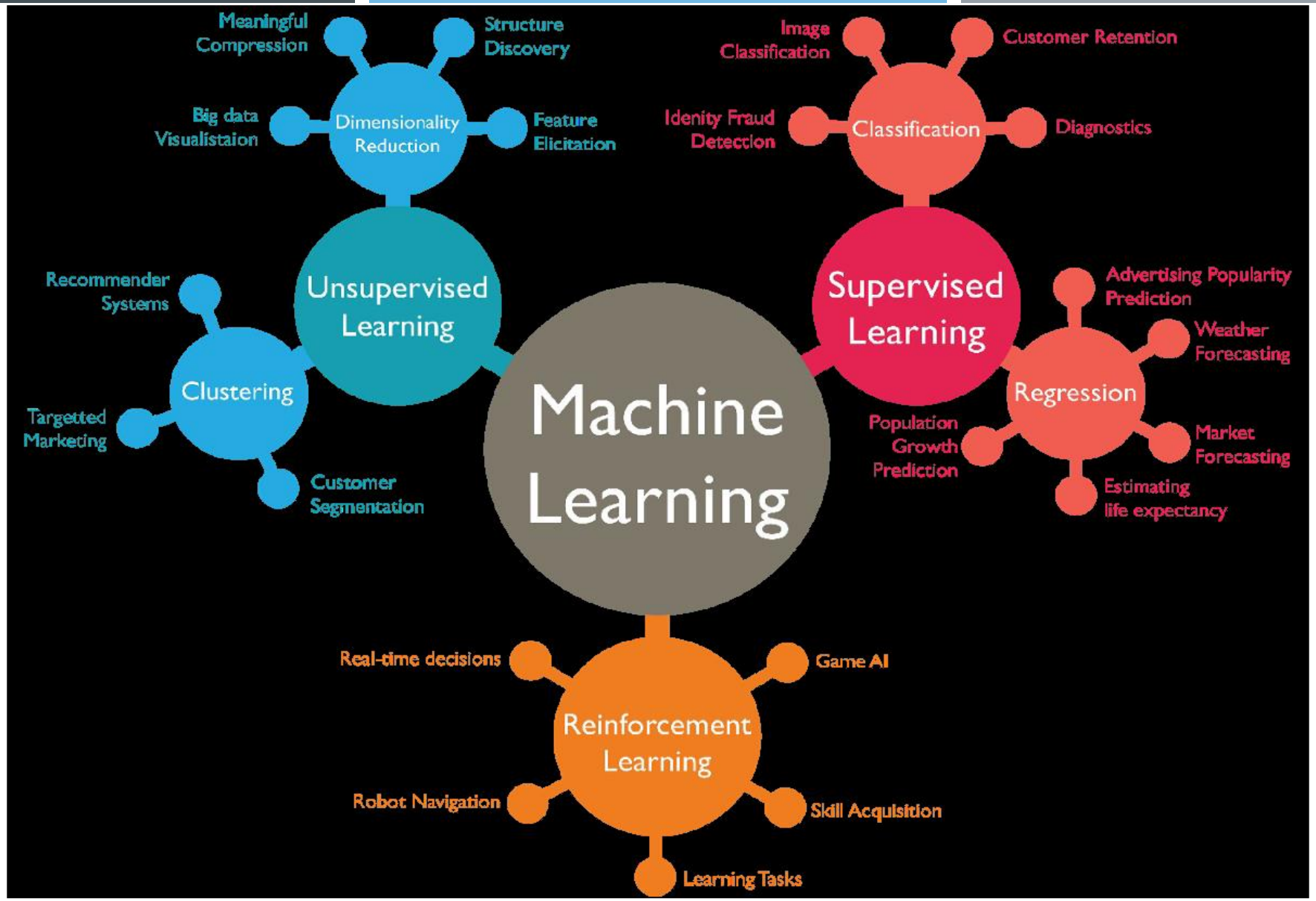
Ability to learn without explicitly being programmed



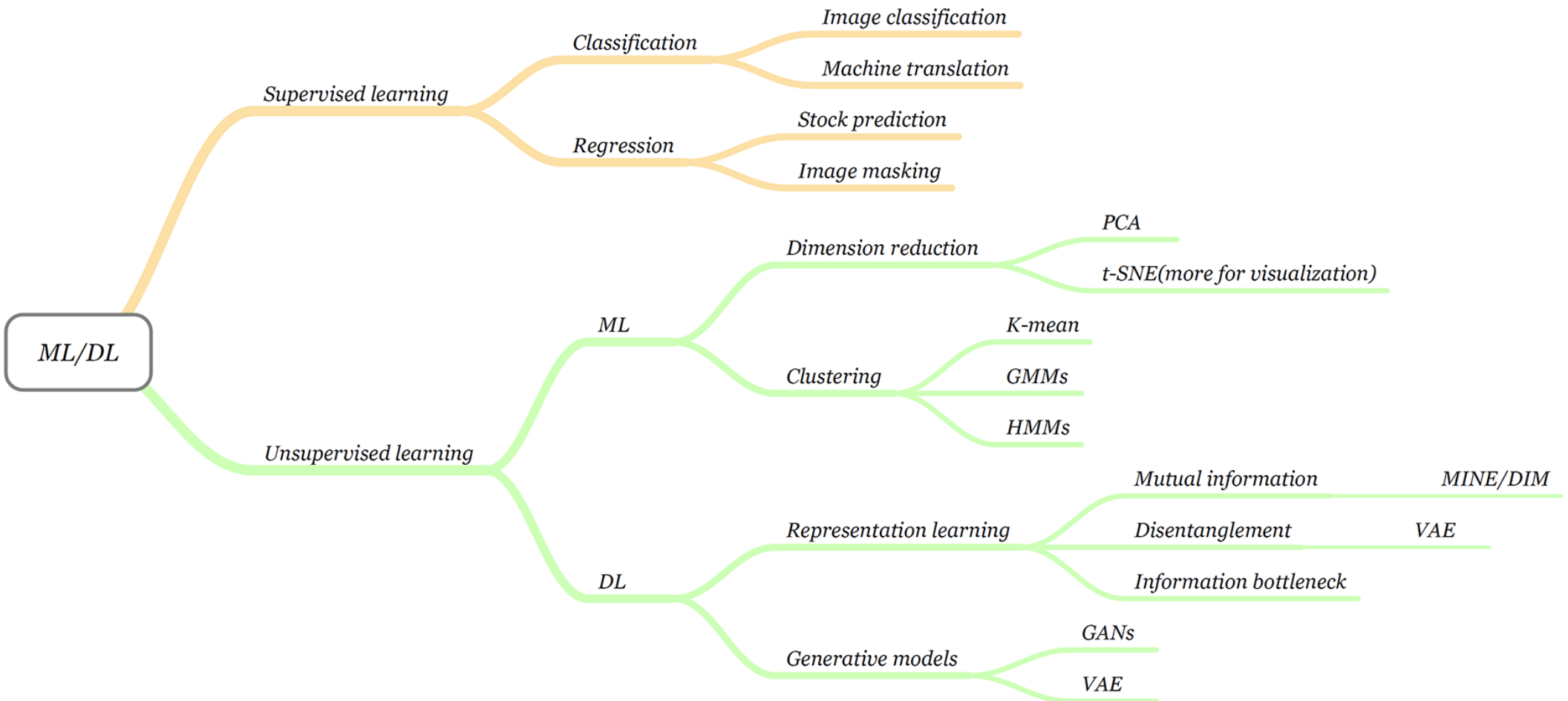
Deep Learning

Extract patterns from data using neural networks





https://medium.com/marketing-and-entrepreneurship/10-companies-using-machine-learning-in-cool-ways-887c25f913c3


ML/DL
Supervised learning
Classification
Image classification
Machine translation
Regression
Stock prediction
Image masking
Unsupervised learning
ML
Dimension reduction
PCA
t-SNE(more for visualization)
Clustering
K-mean
GMMs
HMMs
DL
Representation learning
Mutual information
MINE/DIM
Disentanglement
VAE
Information bottleneck
Generative models
GANs
VAE

SUPERVISED LEARNING

- Prediction of future cases
  - Use the rule to predict the output for future inputs
- Knowledge extraction
  - The rule is easy to understand
- Compression
  - The rule is simpler than the data it explains
- Outlier detection
  - Exceptions that are not covered by the rule.

UNSUPERISED LEARNING

- Learning “what normally happens”
- No output
- Clustering: grouping similar instances
- Example applications: Customer segmentation; Image compression; Learning motifs in Bioinformatics

SUPERVISED LEARNING: PARAMETRIC VS NON-PARAMETRIC

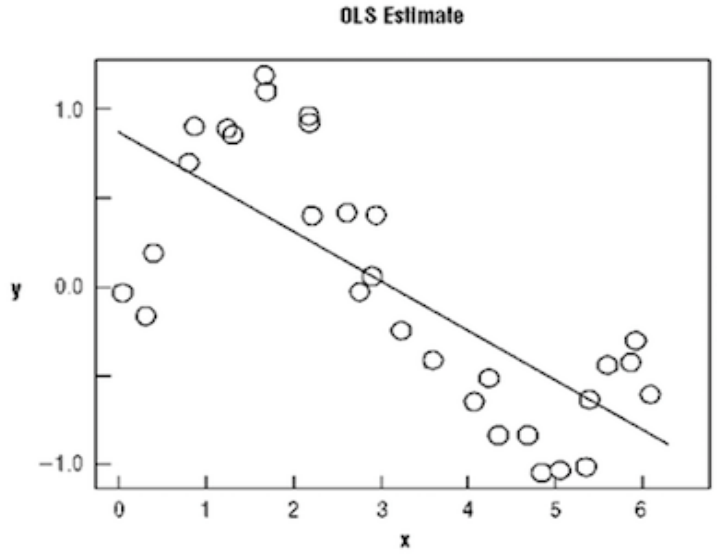
Almost ML models have “parameters” or “weights” than need to be learned.

- Non-parametric model
  - Algorithms that **do not make strong assumptions about the form of the mapping function** are called nonparametric machine learning algorithms.
  - Nonparametric methods are good when you have a lot of data and no prior knowledge (Artificial Intelligence: A Modern Approach, page 757).
  - Seek to best fit the training data in constructing the mapping function, whilst maintaining some ability to generalize to unseen data
- Parametric model
  - Algorithms that simplify the function to a known form are called parametric machine learning algorithms.
  - A learning model that summarizes data with a set of parameters of fixed size (independent of the number of training examples) is called a parametric model. (Artificial Intelligence: A Modern Approach, page 737)
  - A parametric model can predict future values using only the parameters.
  - It requires the function. The parametrics are coefficients of the function.

RESULTS OF PARAMETRIC AND NON-PARAMETRIC REGRESSION

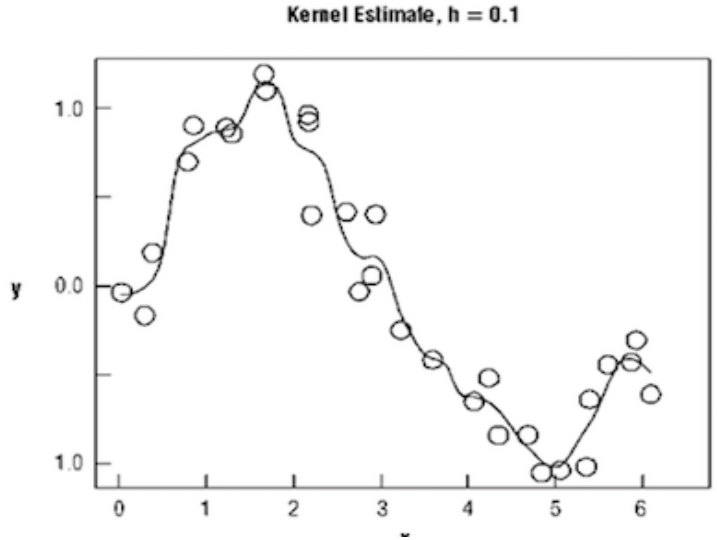
Parametric

- The mapping function as a linear regression line.
- A lot of data points are ignored.



Non-parametric

- Make little to no guesses about the mapping function.
- Greater flexibility and offer a better fit to the data over parametric ones



SUPERVISED LEARNING: PARAMETRIC VS NON-PARAMETRIC

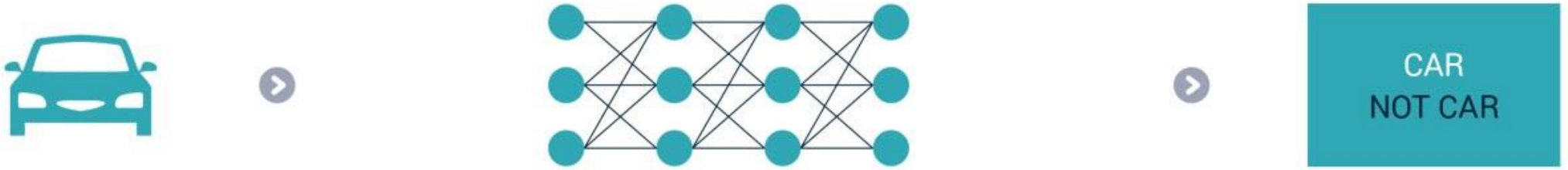
	Non-Parametrics models	Parametric models
Parameters	it simply means that the parameters are (not only) adjustable but can also change. The number of parameters is dependent on the amount of training data. The more training data, the greater the number of parameters.	These parameters are of a fixed-size. The model already knows the number of parameters it requires, regardless of its data. The parameters are also independent of the number of training instances.
Algorithms	k-Nearest Neighbors, Decision Trees, SVM.	Logistic regression, Linear Discriminant Analysis, Perceptron, Naïve Bayes, Simple NN.
Dataset size	Big dataset to achive good performance.	Can generalize with rather small dataset
Learning	Slower to train (have more parameters to consider for the training).	Faster than non-parametric, fewer parameters to train.
Classfication	Very fast	Rather slow. Need to evaluate at least 2 Gaussian on each classification.
Storage space	Very big	Smaller to non-parametric.
Benefits	Performance(more accurate predictions since they offer a better fit to data than parametric ones) . Flexibility, Power.	Simpler, speed, less data.
Limitations	More data, slower, Overfitting	Constrained, Limited complexity, Poor Fit.

Machine Learning

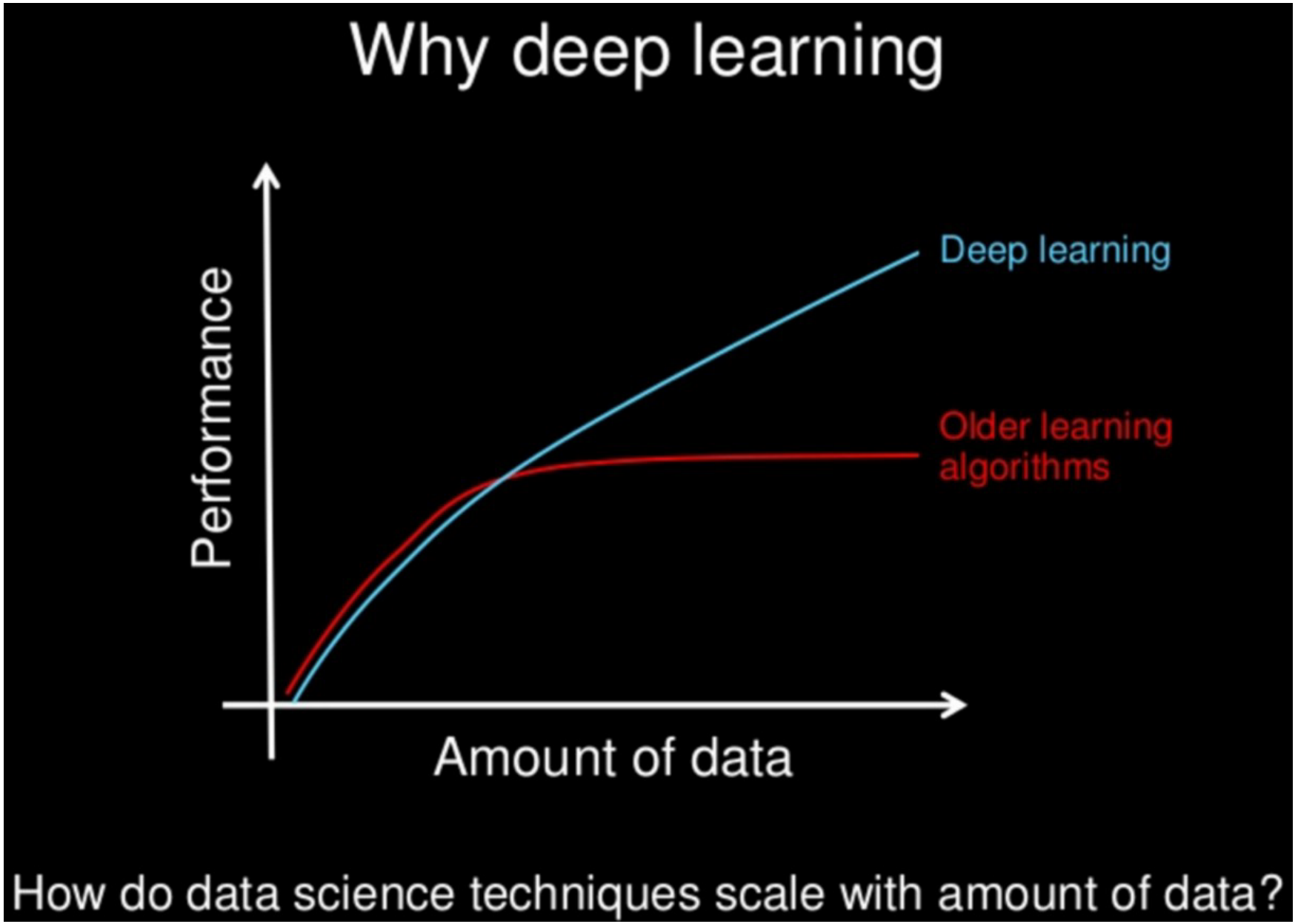


Input → Feature extraction → Classification → Output

Deep Learning



Input → Feature extraction + Classification → Output


Why deep learning
Performance
Deep learning
Older learning algorithms
Amount of data
How do data science techniques scale with amount of data?

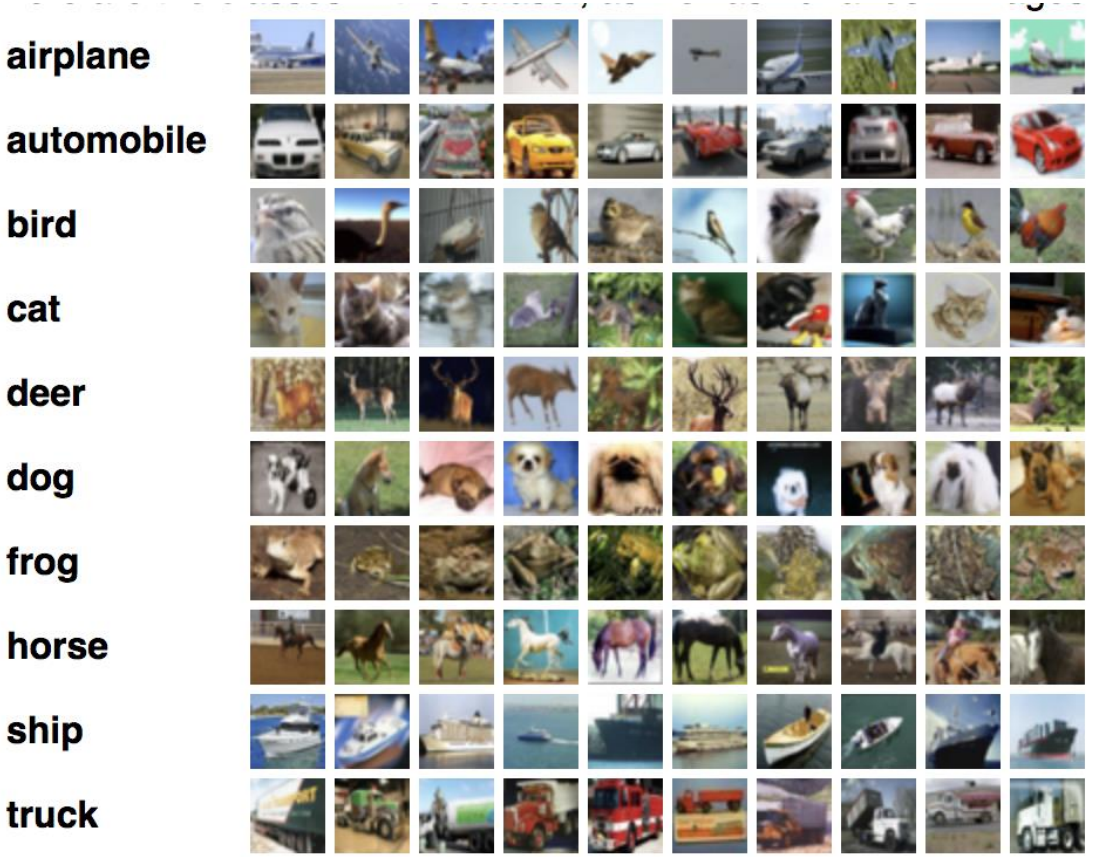
CLASSICAL ML VS. DEEP LEARNING

Classical ML	Deep Learning
Works well for smaller data sets	Improved performance with larger Data
May need feature engineering	Works directly on Raw input
Less Computations	Intensive computations
Low number of parameters	Millions of trainable parameters

SOME POPULAR MACHINE LEARNING CONCEPTS/TERMS

- Supervised Learning
- Unsupervised Learning
- Classification
- Regression
- Decision Trees
- Generative Model
- Discriminative Model
- Deep Learning
- Neural Networks or Artificial Neural Networks
- Reinforcement Learning

IMAGE CLASSIFICATION



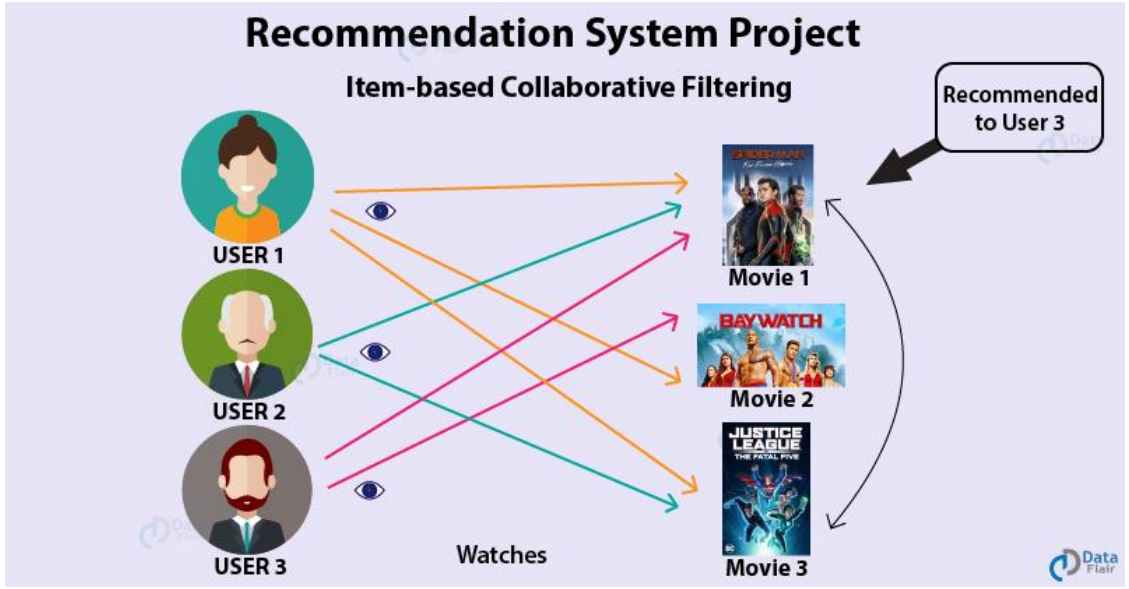
MACHINE LEARNING IN MEDICAL IMAGING



CUSTOMER SEGMENTATION (CLUSTERING)



- Watch: https://www.youtube.com/watch?v=Ep_lDN2hmC0



Recommendation System Generations

XENONSTACK

1st Generation

- Knowledge-based
- Content-Based
- Collaborative Filtering
- Hybrid

2nd Generation

- Matrix Factorization
- Web Usage Mining Based
- Personality Based

3rd Generation

- Collaborative Filtering using DL
- Deep Content based
- Combine Modeling of Users and Items Using Reviews CoNN .etc

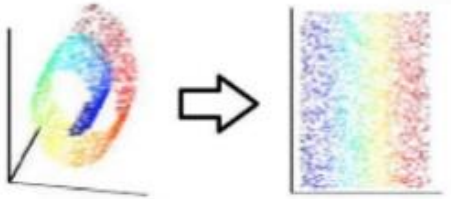
DIMENSIONALITY REDUCTION

Reasons to Reduce Dimensionality

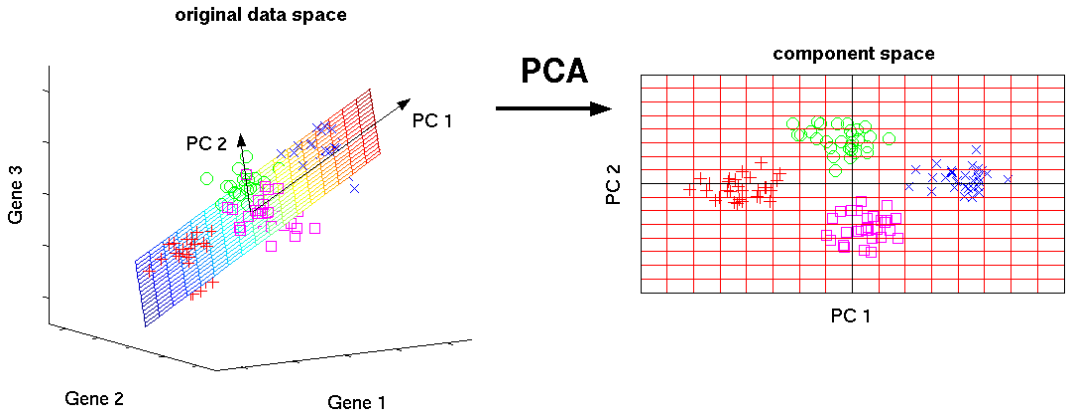
Quicker and more accurate results from machine learning methods

1. Computational issues with large numbers of predictors
2. Inability to use certain statistical methods when not enough observations per predictor exist (such as regression)
3. Less accurate results if the data is noisy (model over-fitting)

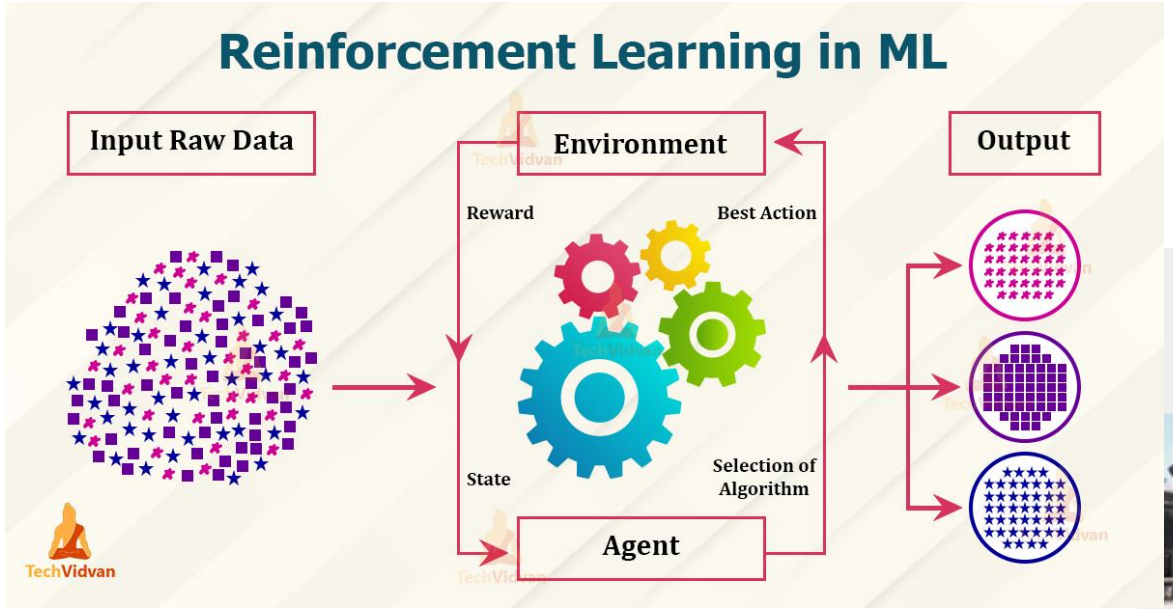
Easier to visualize data

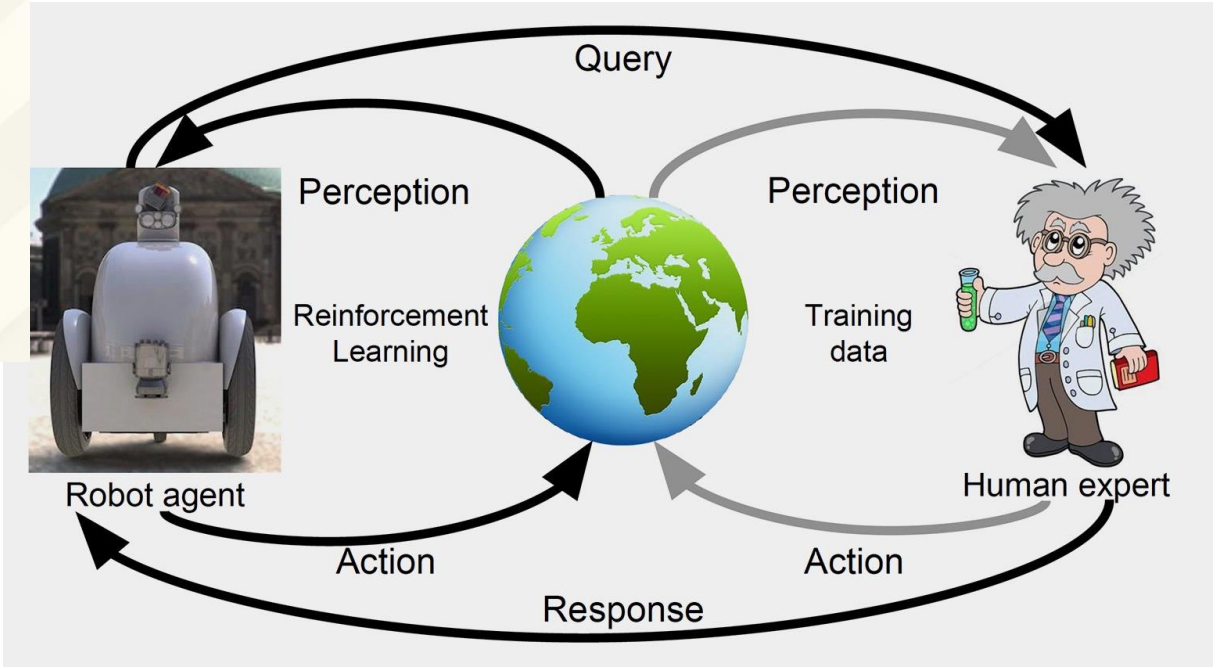


Easier to data mine with fewer predictors

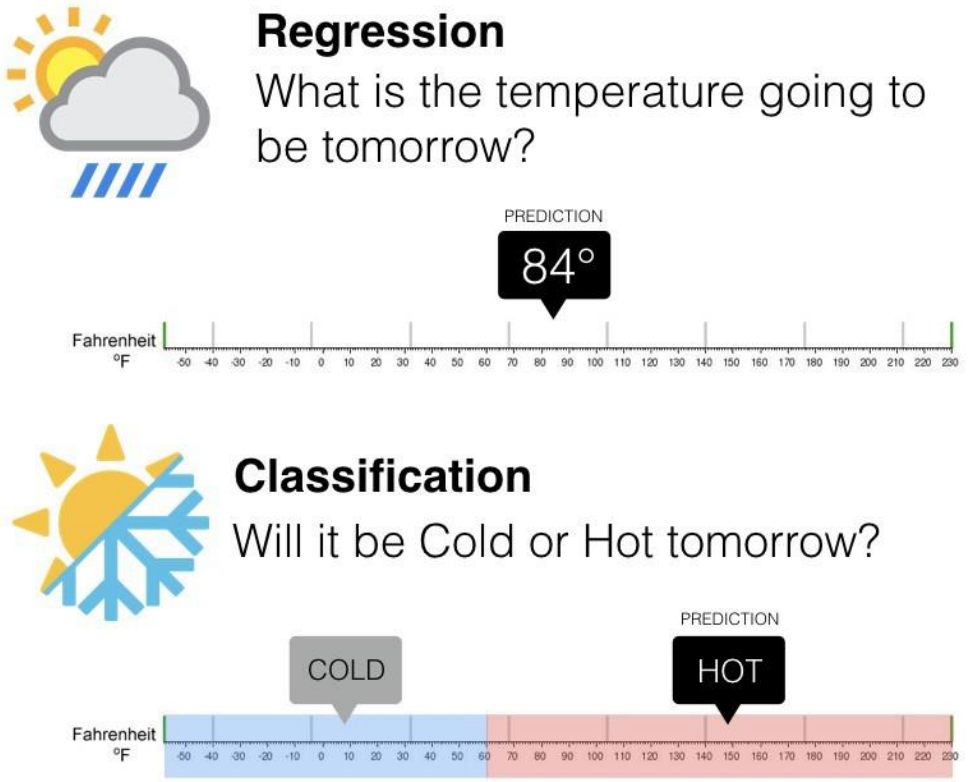


REINFORCEMENT LEARNING

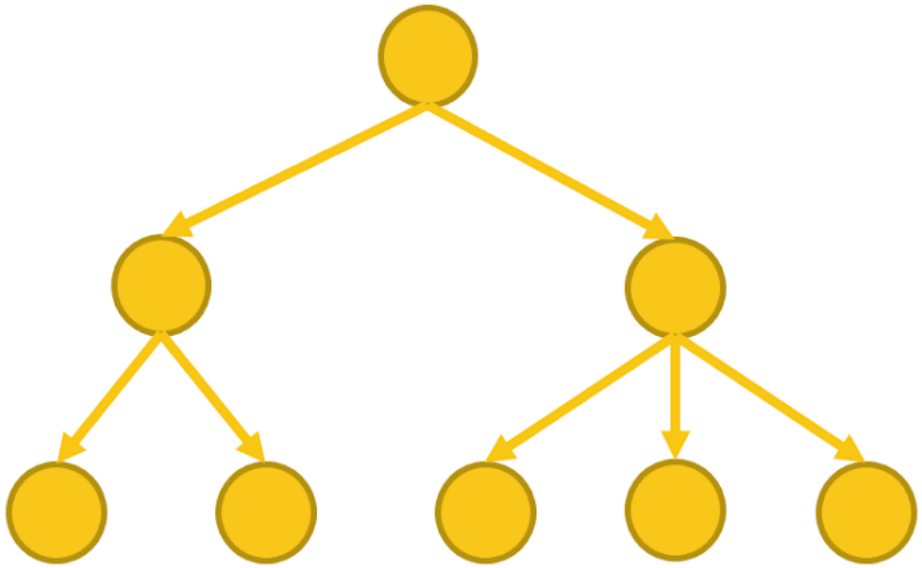




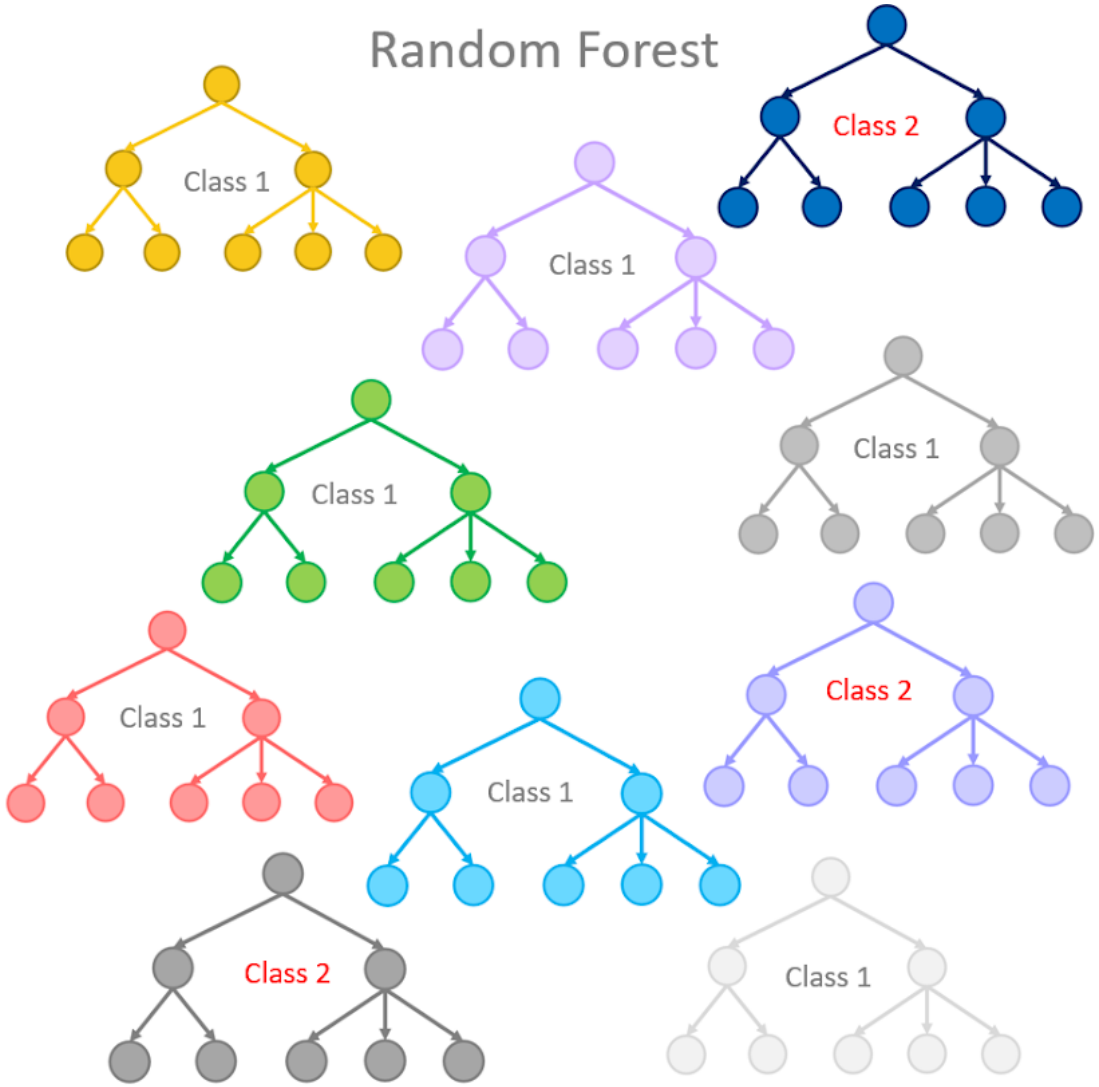
CLASSFICATION VS. REGRESSION



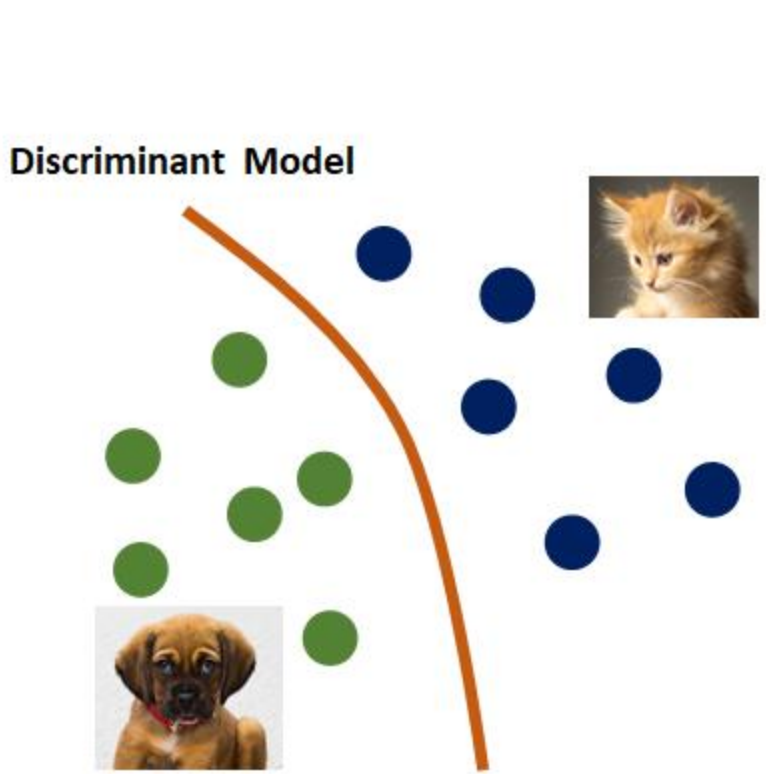
Single Decision Tree

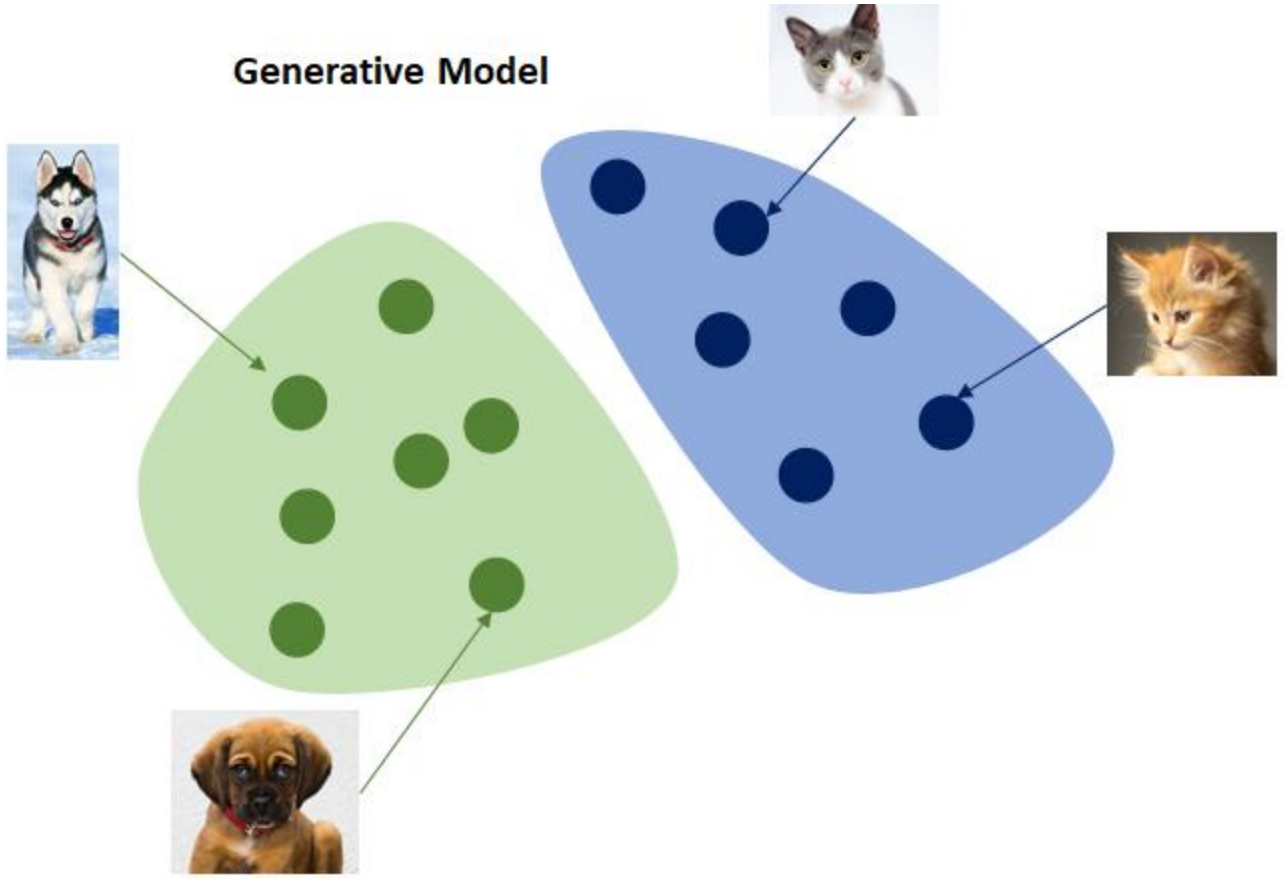


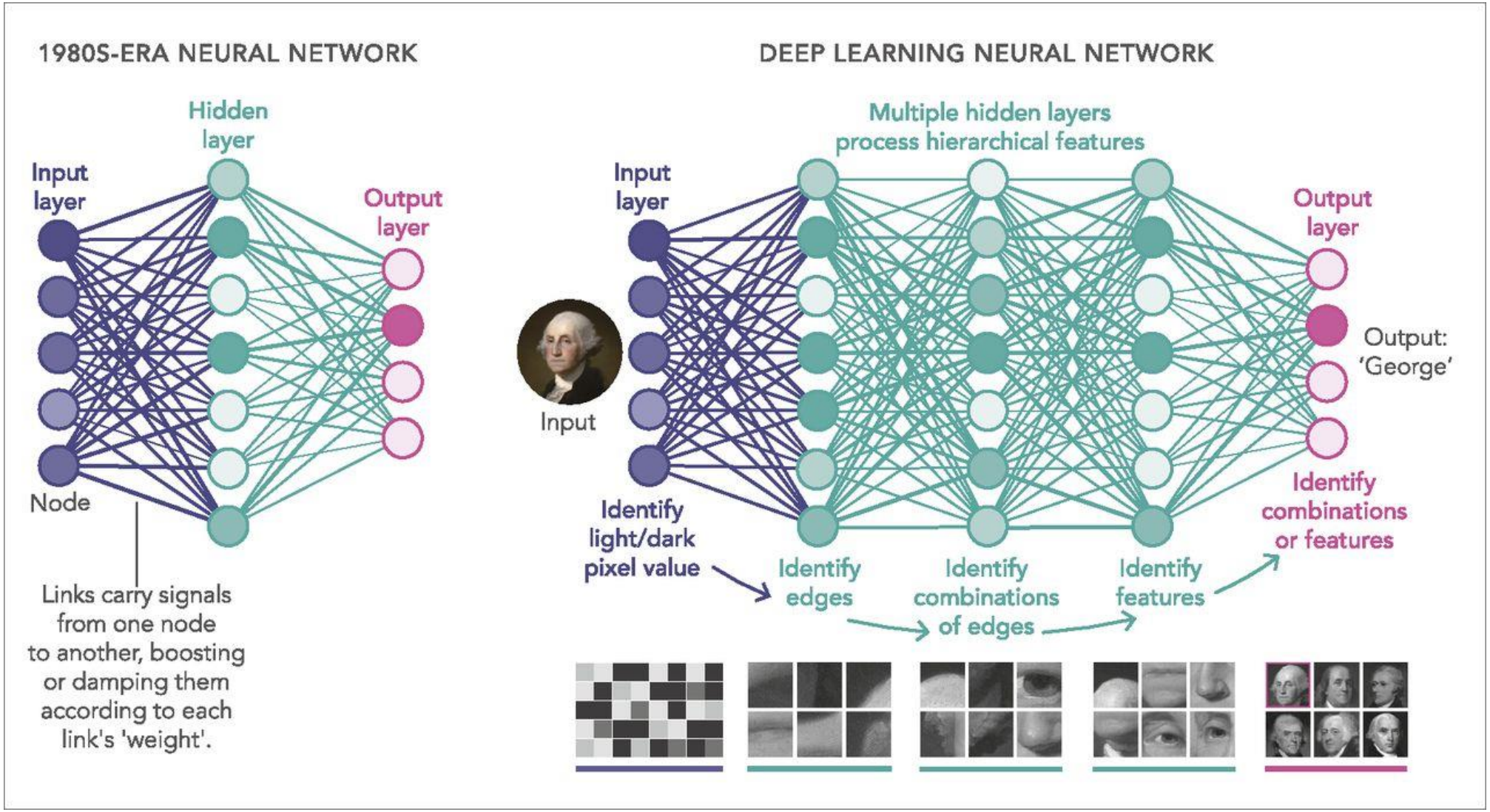
Random Forest



GENERATIVE VS DISCRIMINATIVE






1980S-ERA NEURAL NETWORK
Hidden layer
Input layer
Output layer
Node
Links carry signals from one node to another, boosting or damping them according to each link's 'weight'.
DEEP LEARNING NEURAL NETWORK
Multiple hidden layers process hierarchical features
Input layer
Output layer
Input
Output: 'George'
Identify light/dark pixel value
Identify edges
Identify combinations of edges
Identify features
Identify combinations or features

RESEARCHS IN ML

- https://www.researchgate.net/publication/327547625_A_Research_on_Machine_Learning_Methods_and_Its_Applications
- https://dergipark.org.tr/tr/download/article-file/532680

TOOLS

- Python 3.6 and above
  - Environment:
    - Anaconda, or else.
    - Colab , Kaggle Kernels
    - Jupiter notebooks
- Libs:
  - Pandas, Numpy, Scipy , Matplotlib, Pandas
  - SciKit-Learn
  - Keras(TensorFlow), and maybe Pytorch
- Do yourself a favor and use a Unix like OS.

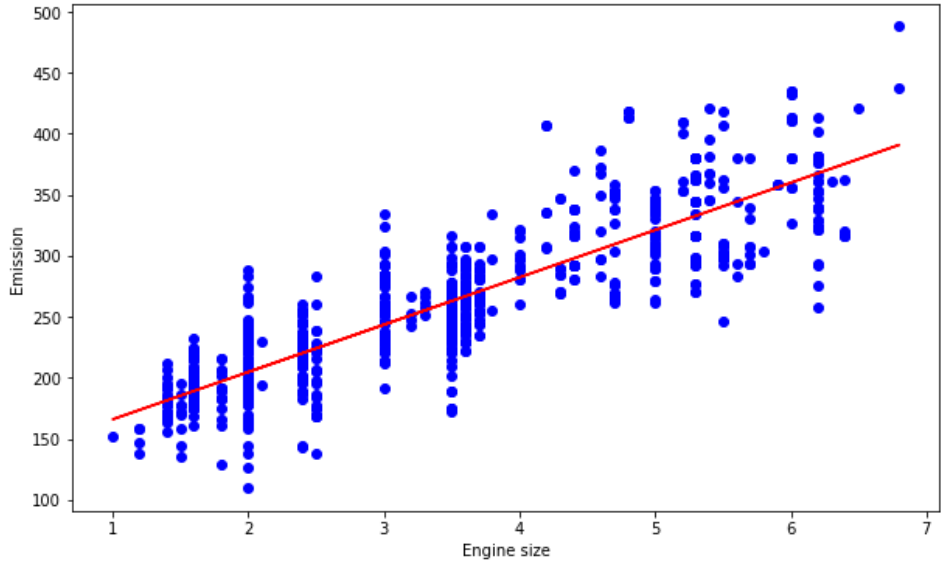
GET STARTED WITH REGRESSION

- Goal: Predict sales of products.
- Linear Regression method.
- Dataset: Vehicles dataset. FuelConsumtionCo2.csv (Softlayer IBM)
- URL: https://medium.com/towards-artificial-intelligence/machine-learning-algorithms-for-beginners-with-python-code-examples-ml-19c6afd60daa

REGRESSION (PREDICTION)

- Linear Regression
- Polynomial Regression
- Exponential Regression
- Logistic Regression
- Logarithmic Regression

LINEAR REGRESSION



- Linear regression is a statistical approach that models the relationship between input features and output. The input features are called the **independent variables**, and the output is called **a dependent variable**. Our goal here is to predict the value of the output based on the input features by multiplying it with its optimal coefficients.

KNOW YOUR DATA

- Identify missing/invalid values
- Understand distributions
  - Data skews and Outliers
- Visualization is the key (master it!)
  - Scatter, box, bar, and histograms plots
  - Heat map, pair, and correlation plots
  - scale axis (log scale)
- The more you know, the better you can build a model!

GET STARTED: EXPLORE AND VISUALIZE DATA

```
# Import required libraries :

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn import linear_model
```

```
# Read csv file :

data = pd.read_csv("Fuel.csv")
data.head()
```

	MODELYEAR	MAKE	MODEL	VEHICLECLASS	ENGINESIZE	CYLINDERS	TRANSMISSION	FUELTYPE	FUELCONSUMPTION_CITY
0	2014	ACURA	ILX	COMPACT	2.0	4	AS5	Z	9.9
1	2014	ACURA	ILX	COMPACT	2.4	4	M6	Z	11.2
2	2014	ACURA	ILX HYBRID	COMPACT	1.5	4	AV7	Z	6.0
3	2014	ACURA	MDX 4WD	SUV - SMALL	3.5	6	AS6	Z	12.7
4	2014	ACURA	RDX AWD	SUV - SMALL	3.5	6	AS6	Z	12.1

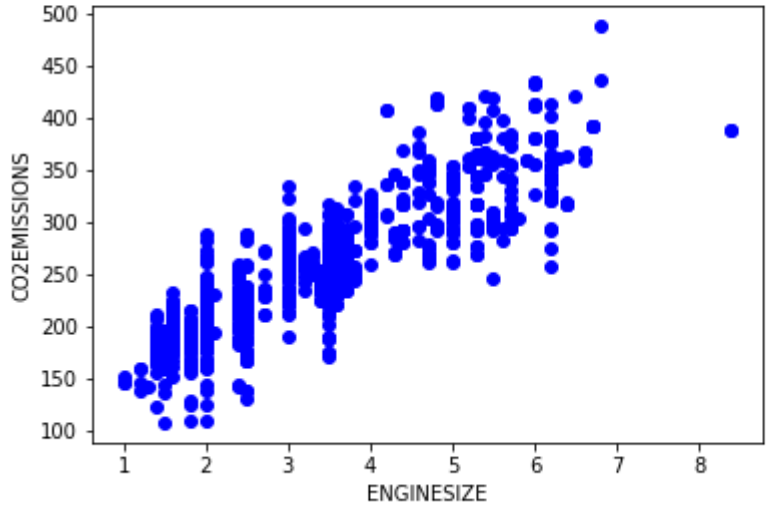
GET STARTED: EXPLORE AND VISUALIZE DATA (CONT)

```
#Let's select some features to explore more :

data = data[["ENGINESIZE","CO2EMISSIONS"]]
```

```
# ENGINESIZE vs CO2EMISSIONS :

plt.scatter(data["ENGINESIZE"] , data["CO2EMISSIONS"] , color="blue")
plt.xlabel("ENGINESIZE")
plt.ylabel("CO2EMISSIONS")
plt.show()
```




```
: # Generating training and testing data from our data :
  # We are using 80% data for training.

  train = data[:(int((len(data)*0.8)))]
  test = data[(int((len(data)*0.8))):]
```

GET STARTED: TRAINING OUR MODEL

```
#Modeling :

#Using sklearn package to model data :

from sklearn import linear_model
regr = linear_model.LinearRegression()

train_x = np.array(train[["ENGINESIZE"]])
train_y = np.array(train[["CO2EMISSIONS"]])

regr.fit(train_x,train_y)


#The coefficients :
print ("coefficients : ",regr.coef_)                    #Slope
print ("Intercept : ",regr.intercept_)                   #Intercept
```

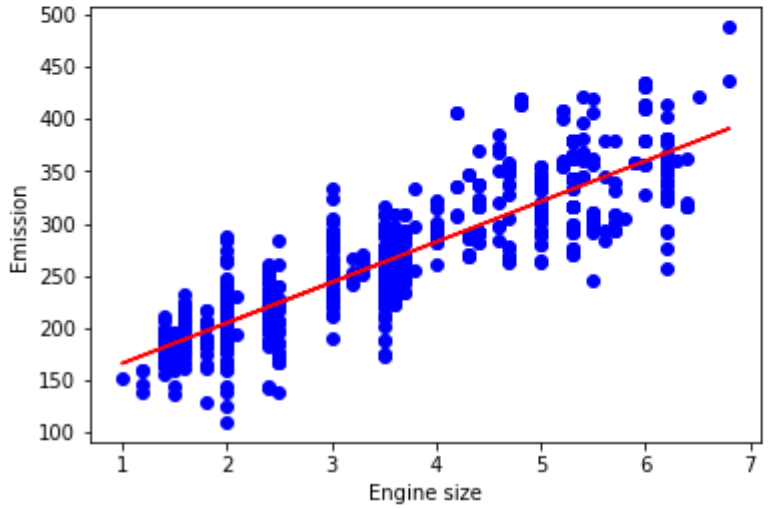
```
coefficients :  [[38.79512384]]
Intercept :  [127.16989951]
```

GET STARTED: PLOT THE BEST FIT LINE

```
# Plotting the regression line :

plt.scatter(train["ENGINESIZE"], train["CO2EMISSIONS"],  color='blue')
plt.plot(train_x, regr.coef_*train_x + regr.intercept_, '-r')
plt.xlabel("Engine size")
plt.ylabel("Emission")
```

```
Text(0, 0.5, 'Emission')
```



GET STARTED: PREDICTION

```
# Predicting values :

# Function for predicting future values :

def get_regression_predictions(input_features,intercept,slope):
    predicted_values = input_features*slope + intercept

    return predicted_values
```

```
# Predicting emission for future car :

my_engine_size = 3.5

estimatd_emission = get_regression_predictions(my_engine_size,regr.intercept_[0],regr.coef_[0][0])
print ("Estimated Emission :",estimatd_emission)
```

```
Estimated Emission : 262.9528329350172
```

GET STARTED: CHECKING ACCURACY FOR TEST DATA

```
# Checking various accuracy

from sklearn.metrics import r2_score

test_x = np.array(test[['ENGINESIZE']])
test_y = np.array(test[['CO2EMISSIONS']])
test_y_ = regr.predict(test_x)

print("Mean absolute error: %.2f" % np.mean(np.absolute(test_y_ - test_y)))
print("Mean sum of squares (MSE): %.2f" % np.mean((test_y_ - test_y) ** 2))
print("R2-score: %.2f" % r2_score(test_y_ , test_y) )
```

```
Mean absolute error: 20.60
Mean sum of squares (MSE): 746.45
R2-score: 0.71
```

VISUALIZE YOUR DATA WITH GOOGLE FACETS

- https://github.com/pair-code/facets
- https://www.youtube.com/watch?v=WVclIFyCCOo

7 STEPS OF ML

- Gathering Data
- Understanding and Preparing Data
- Model Selection
- Training
- Evaluation
- Parameter Tuning
- Making Predictions!
- Watch: https://www.youtube.com/watch?v=nKW8Ndu7Mjw

WHY MACHINE LEARNING IS IMPORTANT

- The nearly limitless quantity of available data, affordable data storage, and the growth of less expensive and more powerful processing has propelled the growth of ML.
- Many industries are developing more robust models capable of analyzing bigger and more complex data while delivering faster, more accurate results on vast scales.
- **Industries that use machine learning:** *Healthcare, Government, Marketing and sales, Transportation, Financial services, Oil and gas, Manufacturing.*
- Read:
  - https://quantilus.com/why-is-machine-learning-important-and-how-will-it-impact-business/;
  - https://www.forbes.com/sites/julianvigo/2019/06/06/why-machine-learning-is-the-future-of-business-culture/?sh=30bd40a32bab
- https://youtu.be/7-qgzaae3ec

THANK YOU